

equivalent delta for identical resistors
will be

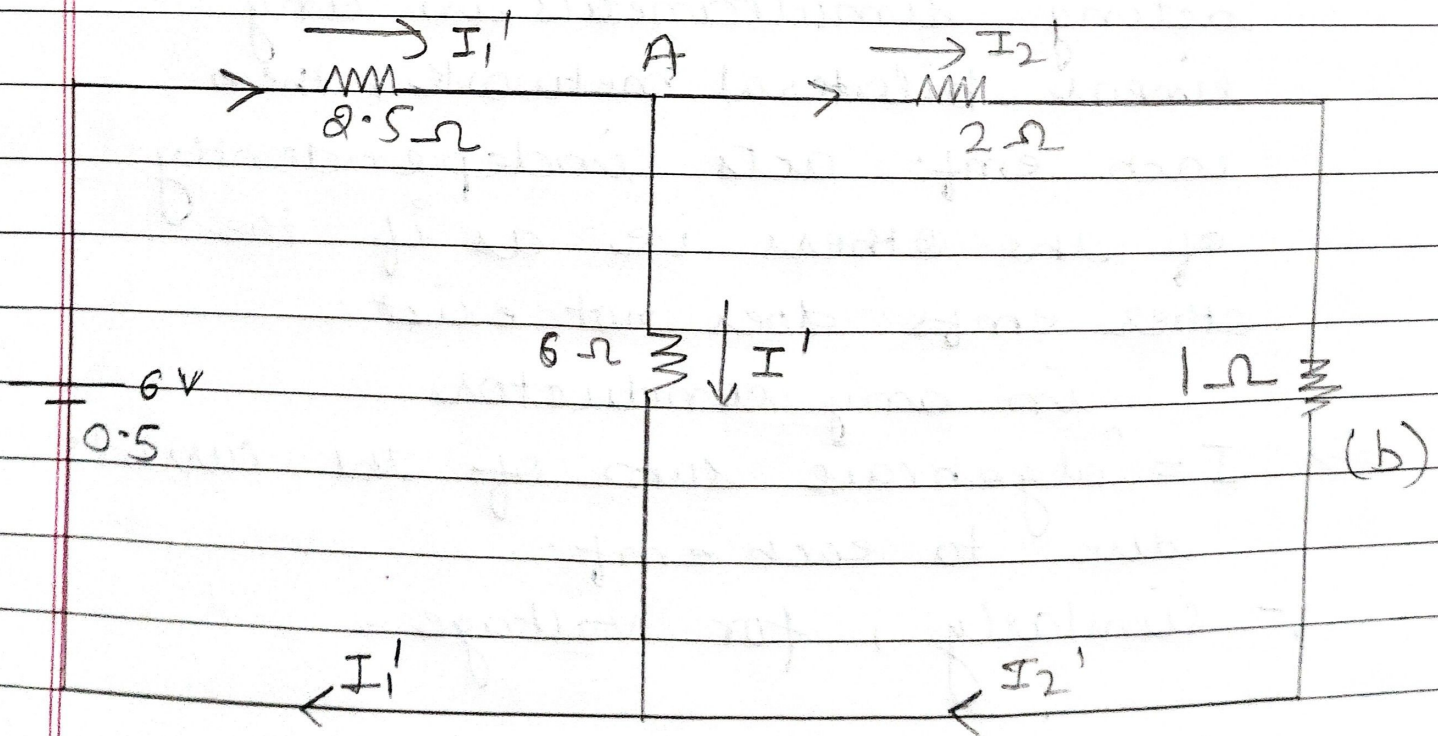
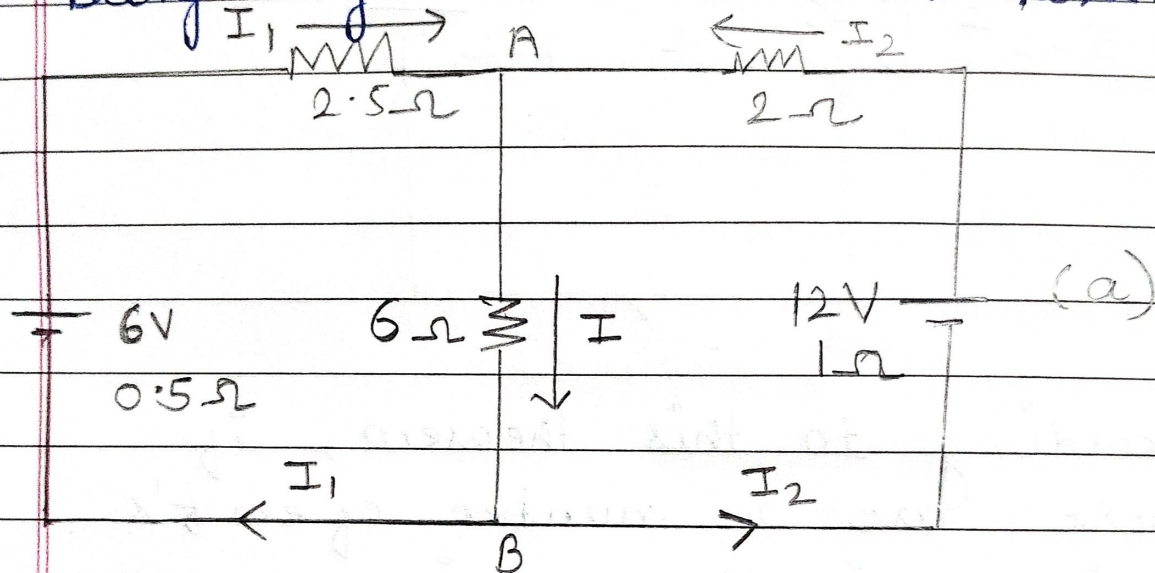
$$R_D = R_S + R_S + \frac{R_S R_S}{R_S} = 3R_S.$$

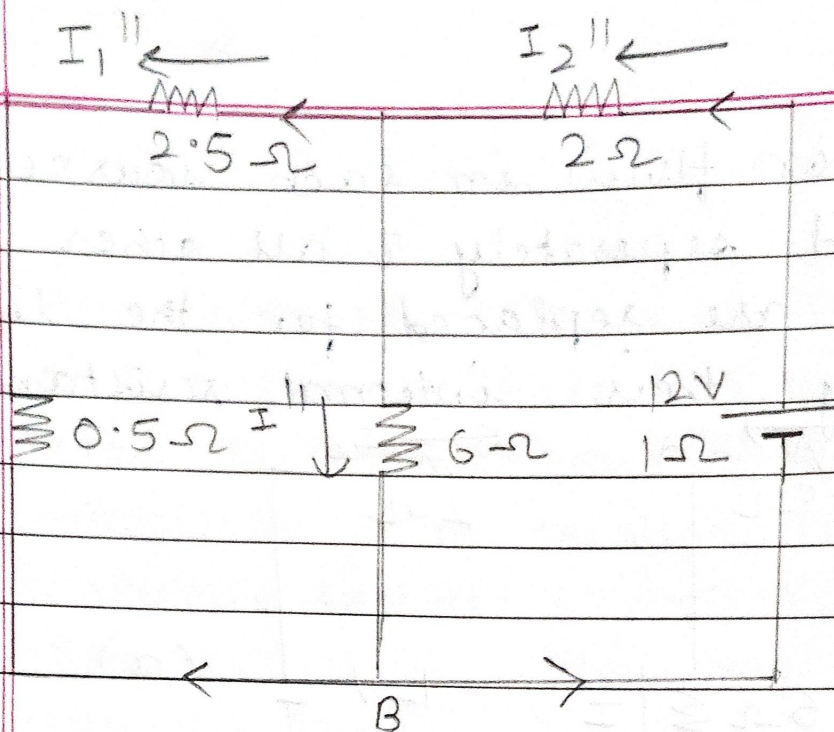


SUPER-POSITION THEOREM

In a network of linear resistances containing more than 1 independent source, voltage across (or current through) an element in a linear circuit is algebraic sum of voltage / or currents through that element which would be

present (or flow) in each source is considered separately & all other sources are replaced for the time being by their internal resistances





⇒ According to this theorem, if there are a number of emf's acting simultaneously in any linear bilateral network, then each emf acts independently of the others i.e. as if the other emfs does not exist.

in any conductor
 ⇒ $I = \text{algebraic sum of the currents due to each emf.}$

∴ similarly, for voltage.

⇒ In other words, current in or voltage across, any conductor of the network is obtained by superimposing the currents &

voltages due to each e.m.f in the network.

* It is applicable only to linear networks where current is linearly related to voltage as per ohm's law:-

Obviously,

$$I_1 = I_1' - I_1''$$

$$I_2 = I_2' - I_2''$$

$$I = I' + I''$$

Thevenin Theorem.

It is a technique for replacing a given network, as viewed from 2 output terminals, by a single voltage source with a series resistance.

→ $V_{AB} = V_{TH}$

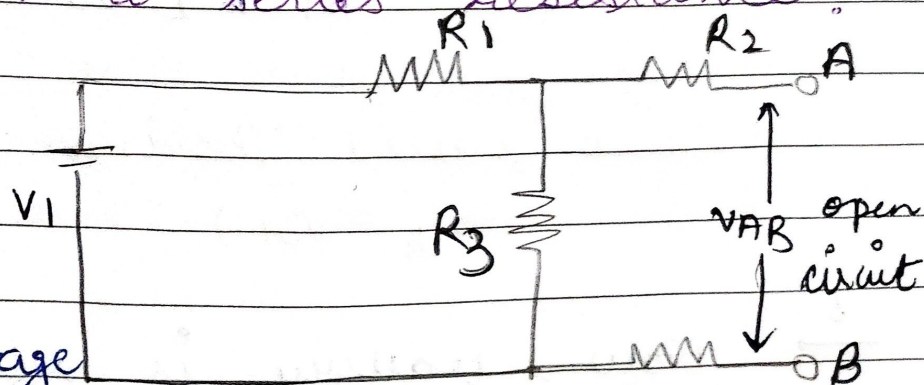
→ $R_{TH}(r) =$

resistance

seen at AB

with all voltage

sources replaced by short circuit & all current sources replaced by open circuit



Steps:-

1 → Remove R_L from terminals A and B and redraw the c't of terminals have become open circuits

2 → Calculate V_{oc} which appears across terminals A & B when they are open i.e. R_L is removed.

$V_{oc} = \text{Drop across}$

$R_2 = IR_2$ where I is c't current when A and B are open

$$I = \frac{E}{R_1 + R_2 + r}$$

$$R_1 + R_2 + r$$

$$\therefore V_{oc} = IR_2 = \frac{ER_2}{R_1 + R_2 + r} \quad [r = \text{internal resistance of battery}]$$

(also called 'Thevenin voltage')

$$V_{Th} = V_{oc}$$

3 → Now, battery is removed from the circuit, leaving the internal resistance (r) behind & redraw

the circuit, when viewed inwards from terminals A & B, the circuit consists of 2 parallel parts one consisting (R_2) & other $(R_1 + x)$.

The equivalent resistance of the network, as viewed from these terminals is given as

$$\begin{aligned} R &= R_2 \parallel (R_1 + x) \\ &= \frac{R_2(R_1 + x)}{R_2 + (R_1 + x)} \end{aligned}$$

It is also called R_{Th} (R_i or R_D) terminals A & B can be reduced to a single source (the whole network) (called Thevenin excluding R_1 source) whose emf equals V_{oc} (or V_{Th}) & whose internal resistance equal to R_{Th} (or R_i)

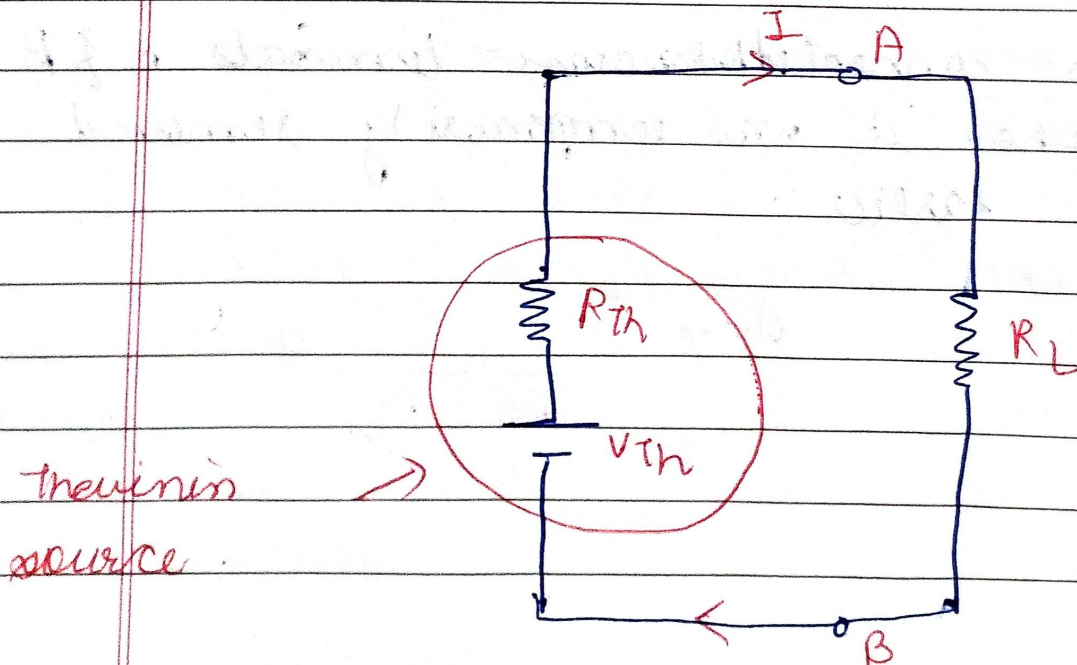
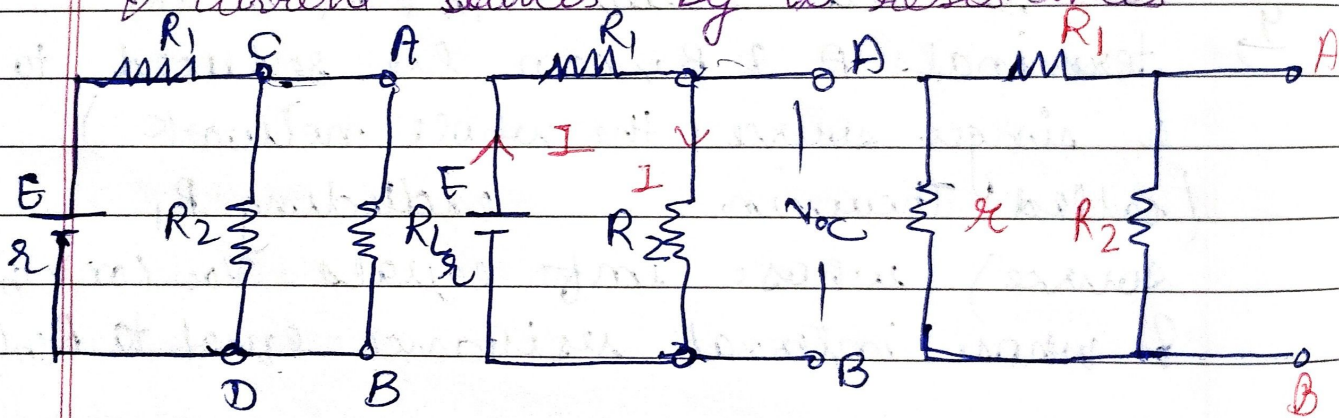
5 →

R_L is connected back across terminals A & B from where it was temporarily removed earlier.

Current through R_L is given by

$$I = \frac{V_{Th}}{R_{Th} + R_L} \quad \text{or} \quad I = \frac{V_{oc}}{R_i + R_L}$$

The current flowing through a load resistance R_L connected across any 2 terminals A & B of a linear, active bilateral network is given by $V_{oc} / (R_i + R_L)$ where (V_{oc} = voltage of open ckt) or (voltage across the 2 terminals when R_L is removed) and (R_i = Internal Resistance of the network) as viewed back into open ckt networks from terminals A & B with all voltage sources replaced by their internal resistance (if any) & current sources by ∞ resistances.



General Instructions for finding Thevenin Equivalent Circuit.

a) When circuit contains Both dependent & independent sources.

(i) The open-circuit voltage V_{oc} is determined as usual with the sources activated or 'alive'.

(ii) The short-circuit is applied across the terminals a & b the value of short circuit I_{sh} is found as usual.

(iii) $R_{Th} = \frac{V_{oc}}{I_{sh}}$

(a) When circuit contains dependent source only

(i) In this case, $V_{oc} = 0$

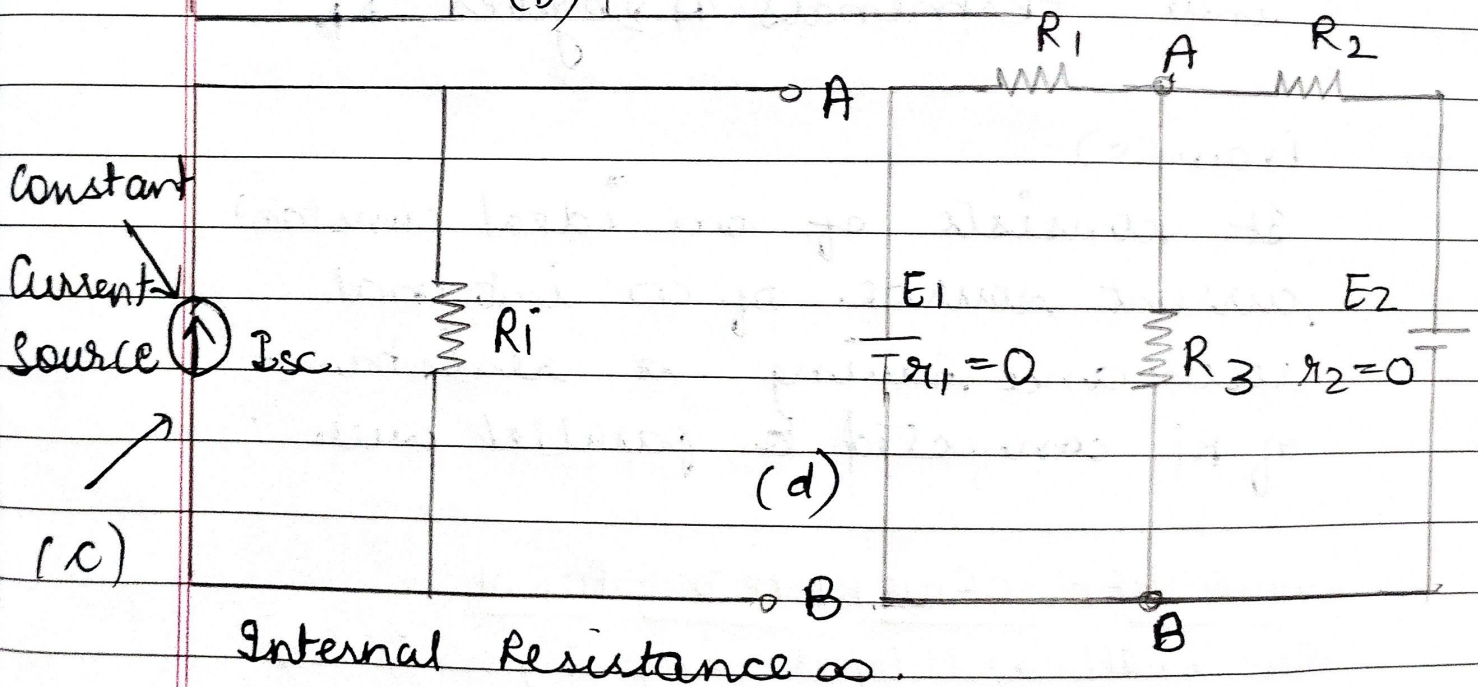
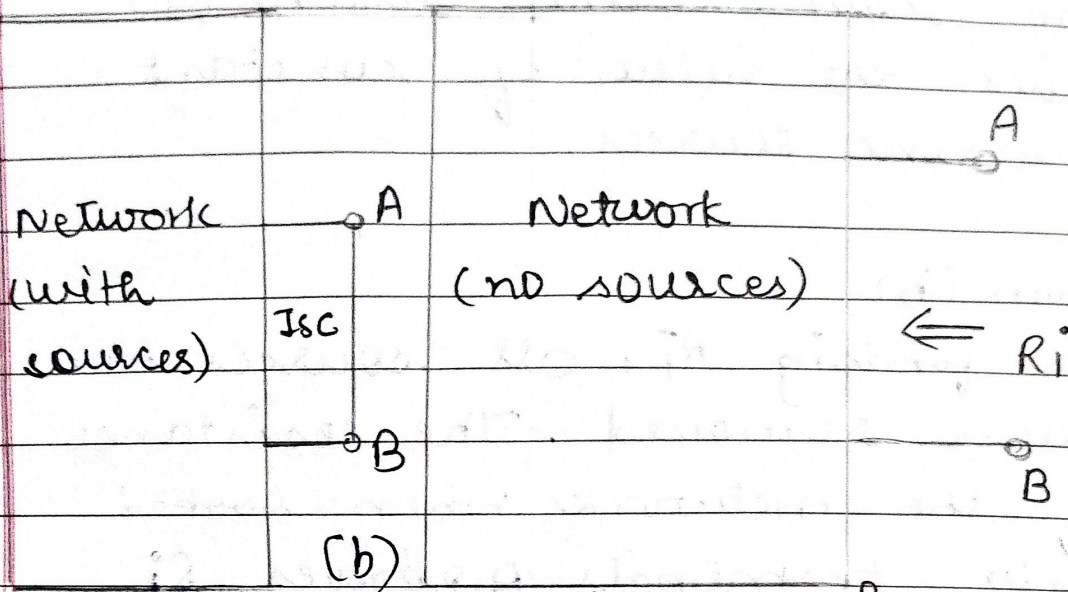
(ii) We connect 1 A source to the terminal A & B & calculate the value of V_{ab} .

(iii) $R_{Th} = V_{ab} / 1 \Omega$

Norton's Theorem

- This is alternative of Thevenin theorem.
- It is dual of Thevenin theorem
- Norton's theorem replaces the network by an equivalent constant current source & a parallel resistance.

* Any 2-terminal active network containing voltage source & resistance when viewed from the output terminals, is equivalent to a constant current source & a parallel resistance. The constant current is equal to the current which would flow in a short-circuit placed across the terminals & parallel resistance is the resistance of the network when viewed from these open c't after all voltage & current sources have been removed & replace by the internal resistance.



Internal Resistance ∞ .

→ from (a)

i) a short is placed across the terminals A & B of the network with all its energy sources present

- The short-circuit current (I_{sc}) gives the value of constant-current sources.
- From (b)
for finding R_i , all sources have been removed. The resistance of the network when looked into terminals A & B gives R_i .
- From (c)
It consists of an ideal constant-current source of ∞ internal resistance having a resistance of R_i connected in parallel with it.

ANOTHER GENERATED FORM :-

The voltage b/w any 2 points in a network is equal to $I_{sc} \cdot R_i$ where I_{sc} is the short-circuit current b/w the 2 points and R_i is the Resistance of the network as viewed from these points with all resistance sources being replaced by their 'internal

resistance (if any) & current sources replaced by open-c'ct.

To find the voltage across R_1 & current through (d)

- If short circuit is placed b/w A & B, then I due to battery e.m.f E_1 is $\boxed{E_1/R_1}$ & due to other $\boxed{E_2/R_2}$

$$I_{SC} = \frac{E_1}{R_1} + \frac{E_2}{R_2} = E_1 G_1 + E_2 G_2$$

$\left. \begin{array}{l} G_1, G_2 \rightarrow \text{Branch} \\ \text{Conductances} \end{array} \right\}$

- Now, the internal Resistance of the network as viewed from A & B simply consists of 3 Resistances R_1, R_2 & R_3 connected in parallel b/w A & B
- here R_3 has not been removed.

$$\frac{1}{R_i} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = G_1 + G_2 + G_3$$

$$R_i = \frac{1}{G_1 + G_2 + G_3}$$

$$V_{Ar_3} = I_{SC} R_i = \frac{E_1 G_1 + E_2 G_2}{G_1 + G_2 + G_3}$$

Current through R_3 is $I_3 = \frac{V_{AB}}{R_3}$

* How to Nortonize a given c't?

1. Remove Resistance (if any) across the 2 given terminals & put a short circuit across them

2. Compute the short c't current I_{sc}

3. Remove the voltage sources but retain their internal resistance, if any. Similarly remove all current sources & replace them by open c't i.e. by ∞ resistance.

4. Next, find the resistance R_1 (also called R_N) of the network as looked into from the given terminals. It is exactly the same as R_{th} .

5. The current source (I_{sc}) joined in parallel across R_1 b/w 2 terminals gives Norton's equivalent c't.